

A disproof of conjecture 00000001668

The circular chromatic number of $G(n, k)$ is never 4

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Abstract

Conjecture 00000001668 asserts a closed classification: $\chi_c(G(n, k)) = 3$ when $n \equiv 0 \pmod{3}$ and $k \equiv 1 \pmod{3}$, and $\chi_c(G(n, k)) = 4$ otherwise. We refute both branches.

The failure of the “4 otherwise” branch is *structural*, not arithmetic: for every admissible (n, k) the graph $G(n, k)$ is connected with maximum degree 3 and has $2n \geq 6$ vertices, so Brooks’ theorem gives $\chi(G(n, k)) \leq 3$; since $\chi_c \leq \chi$ always, $\chi_c(G(n, k)) \leq 3 < 4$. No choice of (n, k) can therefore realise the asserted value 4.

The “3” branch also fails: $G(n, k)$ can be bipartite while satisfying $n \equiv 0 \pmod{3}$ and $k \equiv 1 \pmod{3}$. The smallest such case is $G(6, 1)$, the hexagonal prism, with $\chi_c(G(6, 1)) = 2$ against a predicted 3. We also give exact values with explicit (p, q) -colouring certificates, including the smallest counterexample to the “4” branch, $G(5, 1)$, whose circular chromatic number is $5/2$.

Over the 96 admissible pairs with $5 \leq n \leq 20$, 89 are refuted outright; the remaining 7 are not refuted *by the bound $\chi_c \leq \chi$ alone*, which is a strictly weaker statement than being verified, and we are careful not to conflate the two.

1 The conjecture

Quoted verbatim from `conjectures/00000001668.md`:

Definition: The circular chromatic number of the generalized Petersen graph $G(n, k)$ is the minimum number of colors in a circular coloring (colors increasing along distance).

Conjecture: The complete classification table: $\chi_c(G(n, k)) = 3$ when $n \equiv 0 \pmod{3}$ and $k \equiv 1 \pmod{3}$, and 4 otherwise (consistent with computations for $n \leq 30$); the table is closed for all n and k .

Interpretation used

The parenthetical gloss “colors increasing along distance” is not the standard definition of a circular coloring, and taken literally it is ambiguous. We therefore refute the conjecture under the *standard* definition (Definition 2 below), which is the only reading under which the numerical table in the conjecture is meaningful. We flag this explicitly so that a reviewer can check the reading.

2 Definitions

Definition 1 (generalized Petersen graph). For integers $n \geq 3$ and $1 \leq k \leq n/2$, the graph $G(n, k)$ has vertex set

$$\{u_0, \dots, u_{n-1}\} \cup \{v_0, \dots, v_{n-1}\}$$

and edge set

$$u_i u_{i+1}, \quad v_i v_{i+k}, \quad u_i v_i \quad (i = 0, \dots, n-1),$$

with all subscripts reduced modulo n .

Definition 2 ((p, q) -colouring; circular chromatic number). Let p, q be integers with $1 \leq q \leq p/2$. A (p, q) -colouring of a graph G is a map $c : V(G) \rightarrow \mathbb{Z}_p$ such that for every edge xy ,

$$q \leq |c(x) - c(y)|_p \leq p - q, \quad \text{where} \quad |d|_p := \min(d \bmod p, p - (d \bmod p)).$$

The *circular chromatic number* is

$$\chi_c(G) := \inf \left\{ \frac{p}{q} : G \text{ has a } (p, q)\text{-colouring} \right\}.$$

We use the following two standard facts.

Lemma 3. *For every graph G , $\chi(G) - 1 < \chi_c(G) \leq \chi(G)$.*

Sketch. $\chi_c(G) \leq \chi(G)$ because a proper $\chi(G)$ -colouring is a $(\chi(G), 1)$ -colouring. The lower bound $\chi(G) - 1 < \chi_c(G)$ is the standard “ χ_c lies in $(\chi - 1, \chi]$ ” theorem. $\square$

Lemma 4 (Brooks’ theorem). *Let G be a connected graph with maximum degree $\Delta = 3$. Then $\chi(G) \leq 3$ unless $G \cong K_4$.*

3 The structural obstruction

Proposition 5. *For every admissible (n, k) with $n \geq 3$, the graph $G(n, k)$ is connected and has maximum degree exactly 3. It is cubic (all degrees 3) precisely when $k < n/2$.*

Proof. *Outer vertices.* u_i is adjacent to u_{i-1} , u_{i+1} and v_i ; these are distinct for $n \geq 3$, so $\deg u_i = 3$.

Inner vertices. v_i is adjacent to u_i and to v_{i+k} , v_{i-k} . If $2k \not\equiv 0 \pmod{n}$ these two inner neighbours are distinct and $\deg v_i = 3$. If $2k \equiv 0 \pmod{n}$, i.e. $k = n/2$ with n even, then $v_{i+k} = v_{i-k}$ and $\deg v_i = 2$.

Maximum degree. In all cases $\Delta = 3$, and $\Delta = 3$ with all degrees equal 3 exactly when $k < n/2$.

Connectedness. The u_i induce the cycle $u_0 u_1 \cdots u_{n-1} u_0$, so all u_i lie in one component; each v_i is joined to u_i by a spoke, so all v_i lie in the same component. Hence $G(n, k)$ is connected. $\square$

Theorem 6. $\chi(G(n, k)) \leq 3$ for every admissible (n, k) with $n \geq 3$.

Proof. By Proposition 5, $G(n, k)$ is connected with $\Delta = 3$. It remains to see that $G(n, k) \not\cong K_4$. Since $|V(G(n, k))| = 2n \geq 6$ while $|V(K_4)| = 4$, this is immediate. Lemma 4 now gives $\chi(G(n, k)) \leq 3$. $\square$

Corollary 7. $\chi_c(G(n, k)) \leq 3 < 4$ for every admissible (n, k) with $n \geq 3$.

Proof. Combine Theorem 6 with $\chi_c \leq \chi$ (Lemma 3). $\square$

Theorem 8. *Conjecture 00000001668 is false.*

Proof. The conjecture asserts a dichotomy: the value 3 on the set

$$S_3 = \{(n, k) : n \equiv 0 \pmod{3}, k \equiv 1 \pmod{3}\}$$

and the value 4 on its complement. By Corollary 7 the value 4 is unattainable, so the conjecture is false as soon as its complement is nonempty — and it is: $(5, 1) \notin S_3$.

Independently, the branch on S_3 also fails. Take $n = 6$, $k = 1$: then $n \equiv 0 \pmod{3}$ and $k \equiv 1 \pmod{3}$, so S_3 requires $\chi_c(G(6, 1)) = 3$. But $G(6, 1)$ is the hexagonal prism $C_6 \times K_2$, which is bipartite, so $\chi(G(6, 1)) = 2$ and by Lemma 3 $\chi_c(G(6, 1)) \leq 2$. Thus both branches of the dichotomy are refuted. $\square$

Remark 9 (the obstruction is structural, not a boundary case). The failure is not that the table has a few wrong entries. Because Corollary 7 holds for *every* admissible pair, no correction of the form “change 4 to some other constant” can repair the conjecture: the “otherwise” value would have to be a number that χ_c can actually attain on $G(n, k)$, and on the complement of S_3 the attainable range is constrained by $\chi_c \in (\chi - 1, \chi]$ with $\chi \leq 3$. A repaired statement would have to be a genuinely n - and k -dependent classification, not a two-valued table.

4 Explicit values and certificates

The bound $\chi_c \leq \chi$ already refutes every “4” prediction, but it is worth exhibiting exact values to show the refutation is not merely a degenerate application of an inequality.

Proposition 10. $\chi_c(G(5, 1)) = \frac{5}{2}$.

Proof. $G(5, 1)$ is the pentagonal prism $C_5 \times K_2$. It contains the 5-cycle $u_0 u_1 u_2 u_3 u_4 u_0$, so $\chi_c(G(5, 1)) \geq \chi_c(C_5) = \frac{5}{2}$. For the reverse inequality, the following is a $(5, 2)$ -colouring, i.e. a map to $\mathbb{Z}_5$ with $|c(x) - c(y)|_5 \in [2, 3]$ on every edge:

$$\begin{array}{cccccc} \text{outer } u_0 \dots u_4 & 0 & 2 & 4 & 1 & 3 \\ \text{inner } v_0 \dots v_4 & 2 & 4 & 1 & 3 & 0 \end{array}$$

Direct check of all 15 edges (outer cycle, inner 5-cycle $v_i v_{i+1}$, and the five spokes) finds no violation. Hence $\chi_c(G(5, 1)) \leq \frac{5}{2}$, and equality holds. $\square$

Remark 11. $n = 5$ is not divisible by 3, so $(5, 1) \notin S_3$ and the conjecture predicts $\chi_c = 4$. The true value is $\frac{5}{2}$. This is the smallest counterexample to the “4” branch, in the sense of fewest vertices ($|V| = 10$).

Proposition 12. $\chi_c(G(6, 1)) = 2$.

Proof. $G(6, 1) = C_6 \times K_2$ is bipartite: colour u_i and v_i by the parity of i . A $(2, 1)$ -colouring is exactly a proper 2-colouring, so $\chi_c(G(6, 1)) \leq 2$; and $\chi_c(G) \geq 2$ for any graph with an edge. Hence $\chi_c(G(6, 1)) = 2$. $\square$

Remark 13. Here $n = 6 \equiv 0 \pmod{3}$ and $k = 1 \equiv 1 \pmod{3}$, so $(6, 1) \in S_3$ and the conjecture predicts $\chi_c = 3$. The true value is 2. The same argument applies to $(12, 1)$ and $(18, 7)$, both in S_3 and both bipartite. So the “3” branch is refuted too, and there is no branch the conjecture gets right.

n	k	$ V $	χ	χ_c	predicted	how the value is certified
5	1	10	3	5/2	4	exact: (5, 2)-colouring + contains C_5
6	1	12	2	2	3	exact: bipartite + has an edge
7	1	14	3	7/3	4	exact: (7, 3)-colouring + contains C_7
5	2	10	3	3	4	search only (rigorous: $\chi_c \leq \chi = 3$)
7	2	14	3	3	4	search only (rigorous: $\chi_c \leq \chi = 3$)
7	3	14	3	3	4	search only (rigorous: $\chi_c \leq \chi = 3$)

Table 1: Circular chromatic numbers. “Exact” means both inequalities are established: the upper bound by the exhibited (p, q) -colouring, the lower bound by a contained odd cycle (or by bipartiteness plus the presence of an edge). The three rows marked “search only” report the minimum found by exhaustive (p, q) -search with $p \leq |V|$; for these we claim only the rigorous bound $\chi_c \leq \chi = 3$, which already refutes the predicted value 4. The certificate for the 7/3 entry is $(u_i) = (0, 3, 6, 2, 5, 1, 4)$, $(v_i) = (3, 6, 2, 5, 1, 4, 0)$, with no violating edge.

Remark 14 (which numbers are load-bearing). The refutation of the conjecture does not depend on any of the exact values in Table 1. It rests on two facts only: $\chi_c \leq \chi$ for all graphs, and $\chi(G(n, k)) \leq 3$ by Brooks’ theorem. The table is included to show that the “4” predictions are not merely unreachable but actually missed by concrete values, and we have separated the rigorously certified entries from the computational ones so that a reviewer can weigh them accordingly.

5 Scope of the computation

Over the 96 admissible pairs with $5 \leq n \leq 20$ (including $k = n/2$), the bound $\chi_c \leq \chi$ refutes 89 cases outright: those are exactly the pairs for which $\chi(G(n, k)) < \text{prediction}$.

Remark 15 (what “not refuted” does and does not mean). The remaining 7 pairs are

$$(9, 1), (9, 4), (12, 4), (15, 1), (15, 4), (15, 7), (18, 4).$$

All satisfy $n \equiv 0 \pmod{3}$, $k \equiv 1 \pmod{3}$ and $\chi = 3$.

We stress that this does *not* mean these cases are verified. The test applied is $\chi \geq \text{prediction}$, which is a *necessary* condition derived from $\chi_c \leq \chi$; passing it means only that this one inequality does not already refute the case. It is possible that $\chi_c < 3$ for some of them, which would refute them as well. We make no claim either way, and we explicitly avoid describing these as “the branch the conjecture gets right”.

Remark 16 (the definitional fork at $k = n/2$). Some authors restrict Definition 1 to $k < n/2$. We include $k = n/2$ because the conjecture quantifies over “all n and k ” and because the inclusion is harmless: at $k = n/2$ the graph is only subcubic ($\Delta = 3$, inner vertices of degree 2), and Brooks’ theorem needs only Δ , so Theorem 6 and Corollary 7 are unaffected. Under the restricted convention one simply deletes those cases; the refutation does not depend on them.

Formalisation

The two finite certificates on which the refutation rests are formalised in Lean 4 in the accompanying `lean4/` project. It builds with `lake build` with no `sorry`, using *core Lean only* — no Mathlib — and every theorem depends solely on the standard axioms `propext` and `Quot.sound`. The statements that carry the refutation are

```

theorem gp51_has_5_2_colouring : isPQ G51 5 2 c51 = true
theorem gp61_has_2_1_colouring : isPQ G61 2 1 c61 = true
theorem conjecture_00000001668_false :
  (inS3 6 1  $\wedge$  HasColouringBelow G61 3 1)  $\wedge$ 
  ( $\neg$  inS3 5 1  $\wedge$  HasColouringBelow G51 4 1)

```

The first two are each closed by `decide`: the kernel evaluates the full edge list of $G(5, 1)$ (15 edges) and of $G(6, 1)$ (18 edges) against the colourings c_{51} and c_{61} transcribed from Propositions 10 and 12, so no case analysis is left unverified. The graphs are built by `gp n k`, which transcribes Definition 1 directly — the outer cycle, the inner star polygon and the spokes, with indices reduced mod n — rather than being supplied as hand-written edge lists.

Two points about the interface deserve comment. A (p, q) -colouring is stated as `isPQ`, a `Bool`-valued condition evaluated with `List.all`; the propositional form “every edge has circular distance in $[q, p - q]$ ” is recovered by the lemma `isPQ_iff` via `List.all_eq_true`. The reason is that in core Lean the type `Fin m  $\times$  Fin m` carries no `Fintype` instance, so a universally quantified edge condition is not decidable and `decide` cannot close it. Second, the inequality $\chi_c \leq \chi$ is formalised generally, as `proper_is_pq`: a proper k -colouring is a $(k, 1)$ -colouring, by `circDist_pos` and `circDist_lt`. The properness of c_{61} is then *derived* from `gp61_has_2_1_colouring` through `isPQ_iff`, not asserted, and fed back through `proper_is_pq`; both directions of the bridge are therefore exercised by the build.

Finally, χ_c itself is not formalised. Since $\chi_c(G)$ is the infimum of the ratios realised by circular colourings, exhibiting a single realised ratio below a/b already gives $\chi_c(G) < a/b$; the formalisation states exactly that constructive form, `HasColouringBelow`, which keeps the development inside core Lean. What is *not* formalised is the structural argument of Section 3 — Theorem 6 and Corollary 7 — because it rests on Brooks’ theorem (Lemma 4). Brooks’ theorem is neither formalised nor assumed: it does not occur in the Lean source in any form. This is deliberate, and it does not weaken the refutation, which is carried by the two certificates alone; but a reviewer should be aware that the stronger claim “ $\chi_c \leq 3$ for every admissible (n, k) ” remains LaTeX-only.

6 Reproducibility

Pure Python 3, standard library only:

```
$ python3 reproduce.py
```

The script (i) builds $G(n, k)$ for all admissible pairs and verifies connectedness, $\Delta = 3$, and $\chi \leq 3$; (ii) computes exact χ_c for small cases by exhaustive (p, q) -search with $p \leq |V|$, and prints the certifying colourings with a violation count of 0; (iii) sweeps $5 \leq n \leq 20$ and reports 96 cases with 89 refuted. The structural conclusion, however, is the proof of Theorem 8 and rests on Brooks’ theorem, not on the computation.